

## Experience

### Applied AI Scientist - Applied AI Team

Jul 2026 – Present

#### Inception42, a G42 company · Abu Dhabi, UAE

Machine-Readable Government (MRG) - Policy Intelligence & Compliance

- Building a multi-tenant policy-compliance platform that converts uploaded government policies into machine-readable artifacts and applies them to evidence-backed, reviewable case workflows.
- Engineering bilingual PDF-to-markdown and policy-graph processing with governed extraction review, versioned provenance, explainable corpus-health checks, and controlled publication.
- Developing agentic case workflows spanning evidence retrieval, compliance assessment, human review, version-pinned configuration, and isolated OpenAI Agents SDK execution in Docker and AKS workers.
- Building governed configuration and evaluation lifecycles with customer-scoped versioning, approved ground-truth suites, repeated-run stability analysis, stage-level quality gates, and paired release-regression comparisons.

### Senior ML Engineer

Aug 2024 – Jul 2026

#### Hewlett Packard Enterprise · Singapore

Text-to-SQL AI Platform

- Architected a natural-language query platform for 200+ non-technical users across 75+ tables using LangChain, GPT-4o, schema-aware generation, Redis caching, and database access controls.
- Deployed FastAPI services on HPE Ezmeral Kubernetes with Keycloak SSO, row-level security, Azure AI Foundry model deployments, and team-scoped RBAC.

*Impact: 85% query accuracy · 2,000+ queries/week · 7 business units · query time reduced from 10 minutes to 1 minute*

K8s Watcher - Agentic Monitoring & Remediation

- Designed and deployed an agentic AI system that monitored live Kubernetes state, diagnosed incidents with GPT-4o, and generated context-aware remediation guidance.

*Impact: 50+ incidents diagnosed/week · mean time to resolution reduced by 70%*

Document Planning Hub & OneAI Platform

- Built a LangGraph multi-agent validation service for PDF, PPTX, Excel, and DOCX documents, using specialist agents and an aggregator to produce structured quality reports.
- Led shared AI engineering standards across eight teams, including CI/CD templates, MLOps patterns, Docker conventions, review practices, and an Azure OpenAI on-call assistant.

*Impact: 5,000+ users · document errors reduced by 80% · deployment failures reduced by 60% · time-to-production reduced from 6 weeks to 2 weeks*

Real-Time Address Correction

- Deployed an Azure-hosted LLM service integrating Google Maps API and Azure OpenAI for high-throughput, sub-second delivery-address correction.

*Impact: supply-chain fallout reduced by 35% · manual interventions reduced by 50%*

### Data Scientist - Smart Manufacturing & AI Team

Jan 2022 – Aug 2024

#### Micron Technology · Singapore

- Optimised semiconductor wafer scheduling using reinforcement learning agents (PPO via Ray RLlib) across a 70-machine distributed training cluster, owning design through production deployment
- Deployed agent recommendation pipeline updating production UI every 2 hours; refactored NumPy-based simulation code reducing runtime by 60%
- Fine-tuned an LLM chatbot on 10K+ internal documents (manuals, tickets, product data) for manufacturing knowledge access, achieving BLEU score of 0.82 and 80% first-contact resolution on support queries
- Standardised master data across planning and operations applications, reducing data input and maintenance by 80% and cutting time spent searching for metrics by 50%

*Impact: 0.5% wafer production increase → \$10M annual revenue · 15% reduction in queuing time*

**Dentsu International · Singapore**

- Developed a customer purchase propensity model (85% validation accuracy) deployed to production and integrated into live marketing campaigns
- Deployed a modern data catalog tool to Azure AKS using Terraform, ingesting 10,000+ datasets from Azure Synapse for enterprise data governance
- Built ROAS prediction models and reporting dashboards reducing post-campaign analysis time by 50% and media planning costs by 20%

**Education**

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|----------------------------------------------------------------------------------------------------|---------------------|
| <b>Master of IT in Business</b> (Major: Artificial Intelligence) - Singapore Management University | Jan 2019 – May 2020 |
| <b>Bachelor of Computer Science</b> - University of Petroleum and Energy Studies, India            | Aug 2014 – Apr 2018 |

**Skills**

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**Azure (primary cloud):** Azure AI Foundry, Azure OpenAI Service, Azure ML Studio, Azure Cognitive Services (Speech, Vision), Azure Communication Services, AKS, Azure DevOps, Azure Synapse, Azure Container Apps

**Agentic AI & LLMs:** LangGraph, LangChain, OpenAI Agents SDK, AutoGen, Multi-Agent Orchestration, RAG Systems, LLM Evaluation, Guardrails, GPT-4o, Azure OpenAI, Hugging Face Transformers, Prompt Engineering, LangSmith

**ML & Modeling:** Reinforcement Learning (PPO, Ray RLlib), TensorFlow, Scikit-Learn, XGBoost, Keras, Statistical Modeling, Fine-tuning, SHAP/LIME

**Cloud & Infra:** Kubernetes, Docker, Terraform, Microsoft Entra ID, Keycloak, Istio, Azure Container Apps, AWS (SageMaker, Bedrock, EKS), GCP (Vertex AI, BigQuery), HashiCorp Vault

**MLOps & Deployment:** MLflow, KServe, BentoML, TorchServe, OpenTelemetry, Jaeger, Prometheus, Grafana, GitHub Actions, Azure DevOps CI/CD, Versioned Release Gates, Model Versioning

**Data & Vector Stores:** PostgreSQL (pgvector), Pinecone, ChromaDB, Snowflake, MySQL, Redis, Apache Airflow, Prefect, Kubeflow Pipelines

**Languages & Frameworks:** Python, SQL, Bash, TypeScript, JavaScript/ReactJS, FastAPI, Flask, REST APIs, Microservices, Event-Driven Architecture

**Visualization & Other:** Tableau, PowerBI, Plotly, Streamlit, OpenCV, YOLO, OCR (Tesseract, PaddleOCR), N8N, Microsoft Copilot Studio, spaCy, NLTK

**Certifications & Publications**

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Harvard Business School - Business Analytics · [Innovations in Software Defined Networking and Network Virtualizations \(Publication\)](#)